Product Requirements Document: Indian Smallholder Farmer Application

1. Target User & Field Context

The product is designed for Indian smallholder farmers managing sub-2-hectare plots with mixed crop portfolios. These users possess low-to-moderate text literacy but demonstrate exceptional visual, spatial, and numerical literacy, particularly regarding input costs and market rates. Their core objective is maximizing yield per acre while minimizing upfront cash expenditures on agricultural inputs.

The digital and physical environment imposes strict constraints:

- **Device & Network:** High smartphone penetration, but devices are often shared within households, suffer from cracked screens, and have constantly full storage. Connectivity is strictly intermittent with 4G dropping frequently at the field edge.
- **Field Conditions:** Interfaces must be usable under high sunlight glare, operable with wet or dirty hands, and feature large tap targets with clear offline fallback states. Onboarding must be visually intuitive without requiring precise text data entry.
- **Rhythms & Trust:** Farm workflows are dictated by Kharif and Rabi seasonal sprints. Daily rhythms involve early morning field checks and evening community discussions where peer validation occurs. Trust is built through tangible, field-matched proof and peer success stories; any recommendation requiring upfront capital without assured or insured returns will be rejected.

2. Product Area Requirements

Area 1: Camera-Guided Disease Identification

- **Confidence Level:** Medium (Visual disease detection will bypass input dealers)
- **Problem:** Farmers currently rely on local input dealers (arhtiyas) for visual disease diagnosis, exposing them to biased recommendations heavily incentivized toward expensive chemical interventions.
- **User Need:** Unbiased, accurate crop disease diagnosis and locally actionable, cost-effective treatment advice delivered directly in the field.

- **Functional Requirements:**
    - The UI must explicitly guide the user on framing, lighting, and focus (e.g., "Get closer to the leaf").
    - Provide exact, locally available chemical names and precise dosages in the local language, rather than just naming the disease.
- **Non-Functional Constraints:**
    - **Resiliency & Bandwidth:** Must support asynchronous background uploads for heavy image payloads over poor connections, notifying the user when results are ready.
    - **Field Usability:** UI must have high contrast and large tap targets for use with dirty hands and cracked screens.
- **Success Metrics:** Percentage of successful, readable image captures; rate of treatment adoption versus historical dealer compliance.
- **Open Risks:** Farmers still require physical access to chemicals, leaving the input dealer as a critical bottleneck. Bypassing them entirely may not be feasible.

Area 2: Explainable Yield Forecasts & Visual Scenarios

- **Confidence Level:** Low (Farmers will trust AI yield predictions if localized data is shown)
- **Problem:** Farmers view algorithmic yield predictions as untrustworthy "black boxes." Yield is an emotional metric directly tied to survival, and generic numbers are rejected.
- **User Need:** To understand exactly why a prediction was made, rooted in localized data they can see and verify in their own fields.
- **Functional Requirements:**
    - **Explainable UI:** Break down prediction variables using visual indicators (e.g., "Based on your 20 days of rain, current soil health, and average district yield").
    - **Scenario Planning:** Provide visual sliders for farmers to adjust inputs (e.g., fertilizer amounts) to simulate yield impacts, leveraging their strong numerical literacy.
- **Non-Functional Constraints:**
    - **Shared Devices:** Must allow users to save or compare scenarios locally without consuming significant device storage.
- **Success Metrics:** Engagement time with scenario sliders; user trust scores or qualitative validation in evening community discussions.

- **Open Risks:** High cost-sensitivity means recommendations suggesting more inputs will fail unless paired with assured/insured returns. Trust in AI for something as critical as yield takes multiple seasons to build.

Area 3: Reality-Based Irrigation Advice

- **Confidence Level:** Low (Water management tools will change irrigation habits)
- **Problem:** Farmers currently irrigate reactively based on visual crop stress or whenever power and water happen to be available. Existing weather apps provide generic, folklore-based, or non-actionable forecasts.
- **User Need:** Actionable watering schedules that account for external infrastructure realities and hyper-local weather events.
- **Functional Requirements:**
    - **Reality-Based Scheduling:** Irrigation prompts must strictly align with local electricity availability (e.g., avoiding 2 PM recommendations if agricultural power is only live from midnight to 6 AM).
    - **Hyper-Local Weather Integration:** Tie advice to localized weather APIs. If rain is predicted, explicitly advise delaying irrigation to save pumping costs.
- **Non-Functional Constraints:**
    - **Offline Fallback:** Weather and schedule guidelines must cache effectively during morning network connectivity for offline reference in the field.
- **Success Metrics:** Adherence to delayed irrigation recommendations; estimated pumping costs saved per acre.
- **Open Risks:** Irrigation is heavily constrained by factors outside the farmer's control (canal schedules, erratic grid power). Behavioral change may be blocked by these physical limitations rather than a lack of information.

Area 4: Voice-First Multilingual Advisory

- **Confidence Level:** High (Voice input is preferred over text typing)
- **Problem:** Text-heavy interfaces, complex onboarding, and precise data entry requirements lead to immediate abandonment by users with low text literacy.
- **User Need:** Conversational, localized advice that mimics the dialogue of a trusted human advisor.
- **Functional Requirements:**
    - **Voice-First UI:** Provide a prominent "Hold to Speak" button as the primary input mechanism, avoiding text fields.
    - **Language Switching:** Implement visual or icon-based language selection directly on the main interface, not hidden in settings.

    - **Contextual Memory:** The bot must recall past interactions (e.g., "Did the pesticide I recommended last week work?") to build conversational trust.
- **Non-Functional Constraints:**
    - **Audio Compression:** Must handle voice data efficiently for low-bandwidth connections without exhausting device storage.
- **Success Metrics:** Ratio of voice inputs to text inputs; week-over-week retention of advisory sessions.
- **Open Risks:** Misinterpretation of regional agricultural dialects and colloquial pest names could severely damage trust.

3. Validation Plan

Because this PRD is built on domain heuristics, the following three field tests must be completed before committing engineering resources to production.

1. **Camera Usability in Sunlight**
    - **Method:** Observational Field Study: Ask users to photograph diseased leaves in bright sunlight.
    - **Participant Profile:** Sub-2ha farmers, utilizing older or shared smartphones with cracked screens.
    - **Decision Criteria:** Does the user capture the necessary detail for the AI classification model without significant frustration?
    - **Validates:** Area 1 (camera guidance UI, high-contrast needs, asynchronous upload viability).
2. **Advisory Bot Comprehension**
    - **Method:** Wizard-of-Oz Testing: Users interact with a simulated voice bot using voice notes in local dialects.
    - **Participant Profile:** Farmers communicating exclusively in regional vernaculars (e.g., Malvi, rural Hindi).
    - **Decision Criteria:** Does the system's terminology match the colloquial farming terms (specific pest/disease names) used by the region?
    - **Validates:** Area 4 (conversational memory, voice-first architecture, NLP dictionary requirements).
3. **Electricity/Water Routines**
    - **Method:** Contextual Inquiry & Mapping: Document actual daily routines for water pumping over a one-week period.
    - **Participant Profile:** Farmers reliant on canal schedules and grid electricity for irrigation.

    - **Decision Criteria:** Is an irrigation scheduling feature functionally viable, or are external constraints too rigid for software to influence?
    - **Validates:** Area 3 (reality-based scheduling and the viability of changing irrigation habits).

4. Open Questions & Assumptions

The following low-to-medium confidence assumptions act as foundational risks to the product's success and must be proven true in the field:

- **Will visual disease detection actually bypass the dealer?** (Medium Confidence) The assumption is that giving farmers independent knowledge reduces the dealer's power, but if the dealer is the only source of the physical chemicals, the farmer remains beholden to them.
- **Can localized data displays build trust in AI yield predictions?** (Low Confidence) Yield is tied to survival. It is unproven whether visual scenarios and explainable variables can overcome deep-seated skepticism toward black-box algorithms within a single season.
- **Will software recommendations actually change irrigation habits?** (Low Confidence) We assume farmers will optimize water use if given better data, but erratic power cuts and canal schedules may render even the best software recommendations useless in practice.